This RJ-61 will use the twisted pair cable with a modular 8 connection. See Figure 13.17



Figure 13.17 RJ-14 Ethernet Connector

3. **RJ-21:** The RJ-21 connector has 50 pins with 25 pins at one end and 25 pins at the other end. It is also called as champ connector or Amphenol connector. The Amphenol is a connector manufacturer. The RJ-21 interface is typically used for data communication trucking applications. See Figure 13.18



Figure 13.18 The RJ-21 Ethernet port wire with many other ports on the other end.

13.5 Ethernet Cable Color Coding Techniques

There are three types of wiring techniques to construct the Ethernet cable. It is also known as color coding techniques. They are

- Straight-Through Wiring
- Cross-over Wiring
- Roll-over Wiring

13.5.1 Straight-Through Wiring

In general, the Ethernet cables used for Ethernet connections are "straight-through cables". These cable wires are in the same sequence at both ends of the cable, which means that pin 1 of the plug on one end is connected to pin 1 of the plug on the other end (for both standard – T568A & T568B). the straight through wiring cables are mostly used for connecting PC / NIC card to a hub. This is a simple physical connection used in printers, computers and other network interfaces. See Figure 13.19

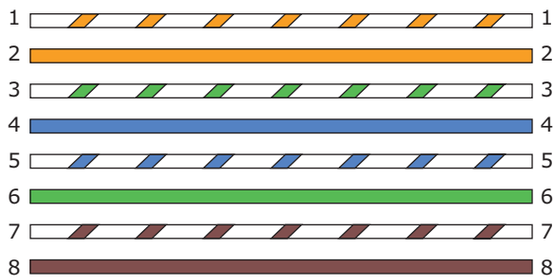


Figure 13.19 Straight through wiring connection

13.5.2 Cross-over Wiring

If you require a cable to connect two computers or Ethernet devices directly together without a hub, then you will need to use a Crossover cable instead. Then the pairs(Tx and Rx lines) will be crossed which means pin 1 & 2 of the plug on one end are connected with pin 3 & 6 of the plug on other end, and vice versa (3 & 6 to pin 1 & 2).

The easiest way to make a crossover cable is to make one end to T568A colour coding and the other end to T568B. Another way to make the cable is to remember the colour coding used in this type. Here Green set of wires at one end are connected with the Orange set of wires

XII CompApp_English version CHAPTER 13.indd 187 23-02-2020 13:00:29